



FUSION: AI-Powered Spatial Multi-Omics x Histopathology Analysis Platform

High-resolution histopathology + spatial omics + machine learning in one platform

ISMB 2026 • FUSION Demo Session

University of Florida
Computational Microscopy Imaging Laboratory
Funding: NIH U54DK134301 and OT2OD033753 (HuBMAP)



FUSION Demo Session

01 **Why Fuse Histology + Spatial Omics**

02 **What is Fusion? How it is different compared to other OMICS tools?**

03 **The FUSION Analysis Pipeline and Supported Data Type**

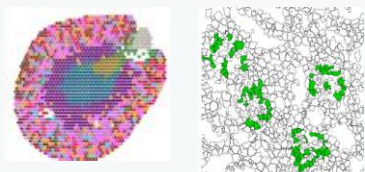
04 **Live Fusion Walkthrough via Fusion JupyterHub Interface**

05 **Heading Next: FUSION 2.0**

Why Fuse Histology and Spatial Omics?

Omics tells you what molecules are present. Histology tells you the morphological properties of Functional Tissue Units (FTU).

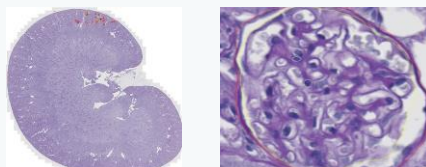
Objective: Combined processing of both modalities and perform analysis, per FTU.



Spatial Omics

Molecular readout

- Transcript counts per spot / cell
- Reference-atlas deconvolution
- Cell-type proportions



Histopathology

Structural readout

- FTU shape & basic morphology
- Quantitative pathomic features
- Gold-standard clinical context



Machine Learning

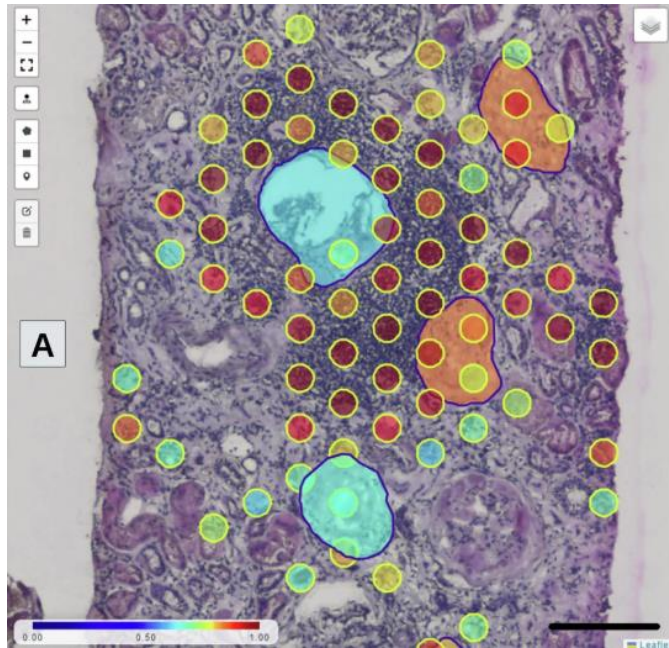
The bridge

- Deep-learning FTU segmentation
- Overlap-weighted spatial aggregation

FUSION's mission: combined record per FTU— morphology + molecular composition.

What Is FUSION: The Core Idea

A one-stop shop for typically siloed analyses - morphology and molecular biology of FTUs, in one platform.



High-Resolution AI-Based Histopathology + Spatial Omics

- Anchored to segmented structures, not arbitrary bins/grids
- Morphology and omics in a joint record and unified processing
- Every processing step: a reusable, versioned plugin
- Executes in cloud-HPC (no user infra set up needed)
- Availability of open-sourced pre-existing data or ingest user data

What Do You Need to Know to Run FUSION?



Whole Slide Image

H&E or PAS stain. Frozen or FFPE tissue, high resolution.
40X magnification and 0.25 MPP preferred.

+



Paired Spatial Omics Assay

Seurat v5 (.RDS) or Scanpy (.h5ad) — same tissue section.
(Not direct sequencing input like raw FASTQs)

MCS

Multi-Compartment Segmentation

Segments histopathological FTU compartments

PFE

FTU Pathomic Feature Extraction

Extracts morphology per FTU

Cell Type Annotation

Label Transfer + Spot Annotation

Cell-type proportions / spot

FTU Aggregation

FTU-Level Molecular Aggregation

Spatio-molecular composition
mapped to individual FTUs

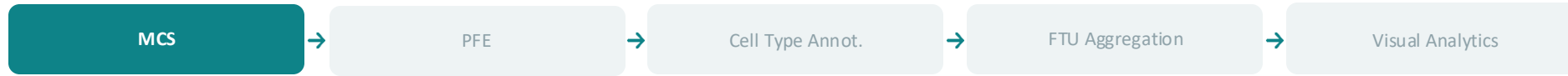
Visual Analytics

Visualization & Analysis

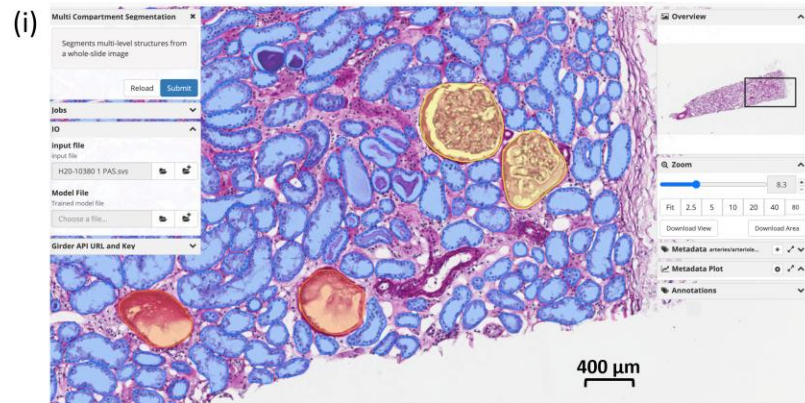
Interactive plots, lasso select,
custom analysis code execution

A flexible plugin architecture: raw WSI + molecular profiles -> a structure-aware, multi-omic feature set.

Multi-Compartment Segmentation (MCS): The Anchor



- Deep-learning based AI auto-segmentation: trained model across diverse kidney disease types and stains
- Fully automated analysis pipeline — no manual annotation required

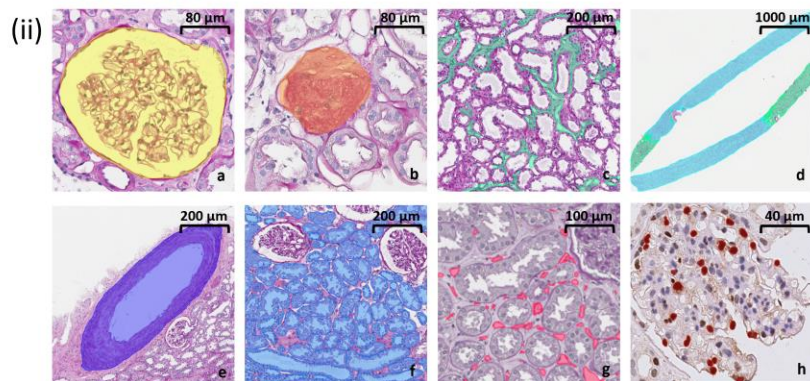


INPUT

High-resolution H&E- or PAS-stained WSI

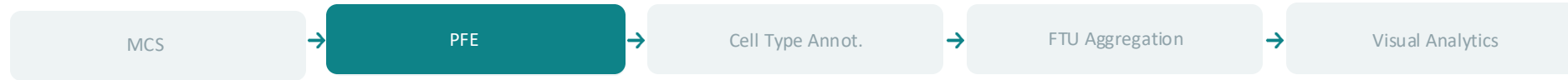
OUTPUT

Segmented FTU structures with annotation boundaries written in a downloadable JSON file

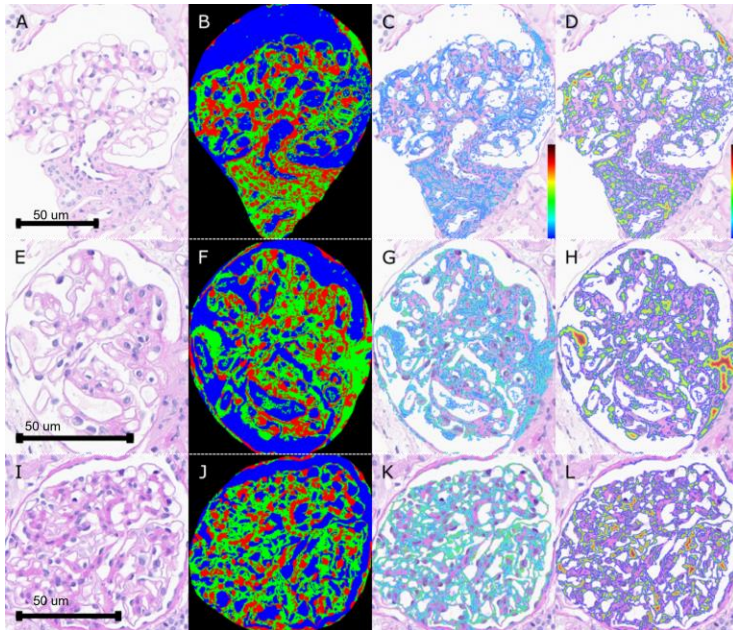


AI generated renal FTU segmentation in high magnification. a: Non-Globally Sclerotic Glomerulus; b: Globally Sclerotic Glomerulus; c: Cortical Interstitial space; d: Cortical (green) and Medullary (blue) areas; e: Artery; f: Cortical tubules; g: Peritubular Capillaries; h: Podocyte nuclei.

FTU Pathomic Feature Extraction (PFE)



- FTU-aware 72 features: morphology, texture, color, spatio-nuclei organization
- Computed across 3 sub-compartments per FTU: nuclei, PAS+ (cytoplasm/matrix), and lumen



INPUT

High-resolution H&E-stained WSI with segmented FTUs

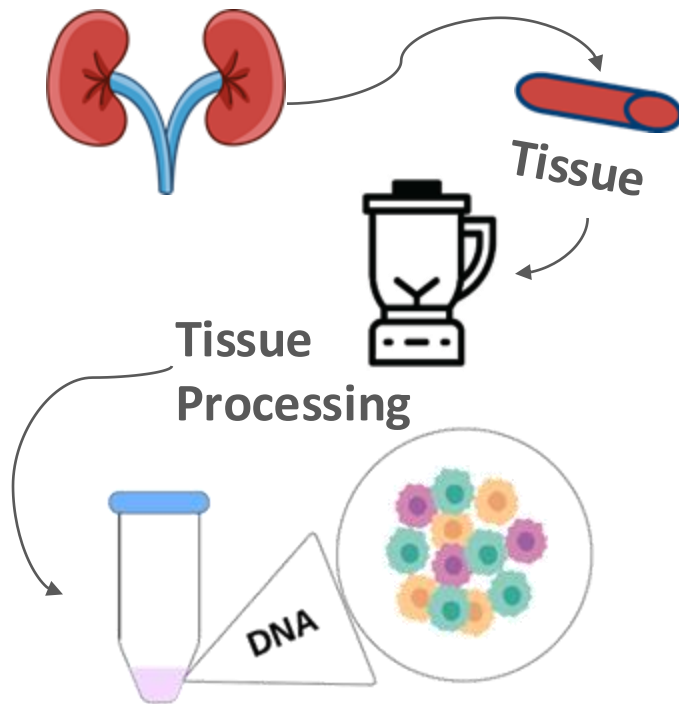
OUTPUT

Per-FTU pathomic feature table in csv (72 features), downloadable, exportable and linkable to omics data

Segmented (A, E, I) and sub-compartmentalized (B, F, J: nuclei red, PAS+ green, lumen blue) glomerular masks from different stages of disease progression: (C, G, K) standard deviation of PAS+ color mask, smoother color differences between glomerular capillary and lumen indicative of a normal healthy glomerulus. (D, H, L) mapped distance transform indicating fibrotic/sclerotic pathways inside glomerulus with greater PAS+ thickness.

Image sources: Sarder et al. 2026

Background: Bulk and Single Cell Omics



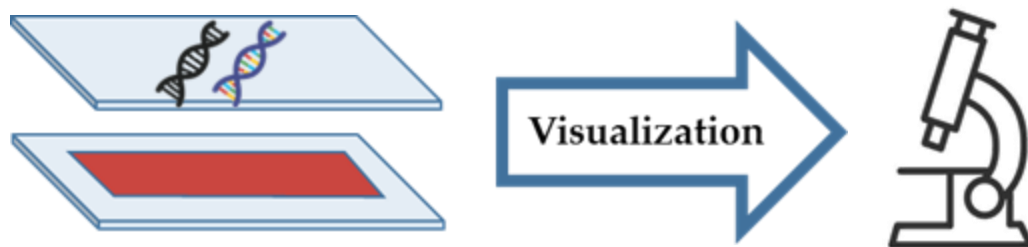
- Learn molecular information from dissociated tissues
- Loss of the spatial dimension
- Cellular neighborhoods NOT preserved

Background: Spatial Omics

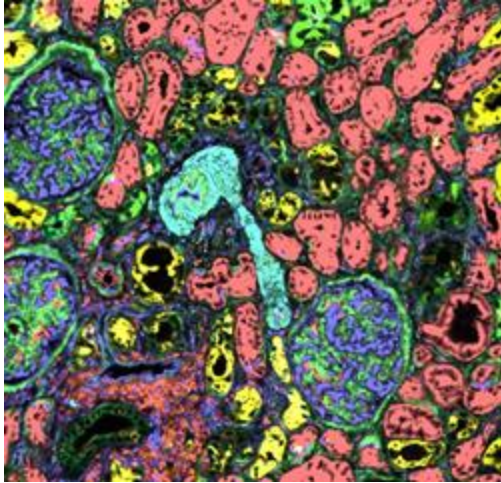


- Measure molecular information from in-tact tissue
- Spatial dimension added to data
- Preserves cellular neighborhoods

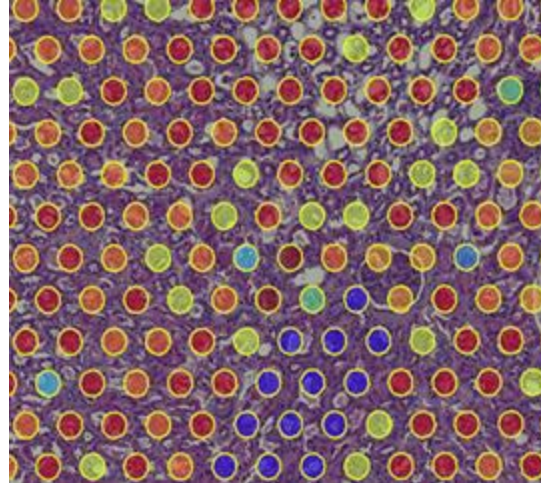
Same- or adjacent-section Spatial -Omics



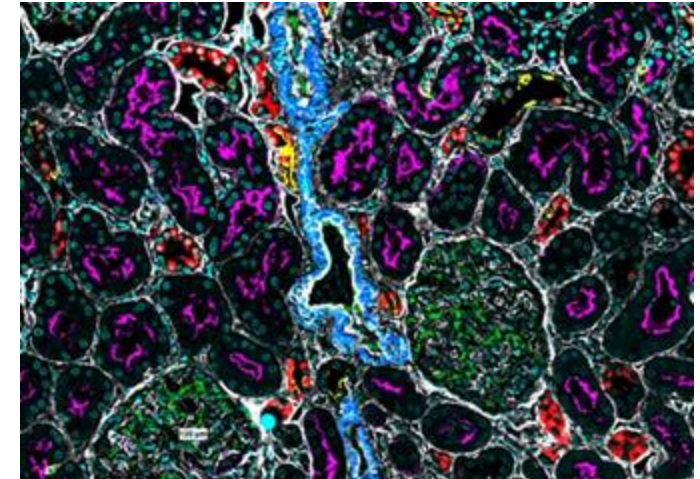
Spatial Omics: What's Underneath the Surface?



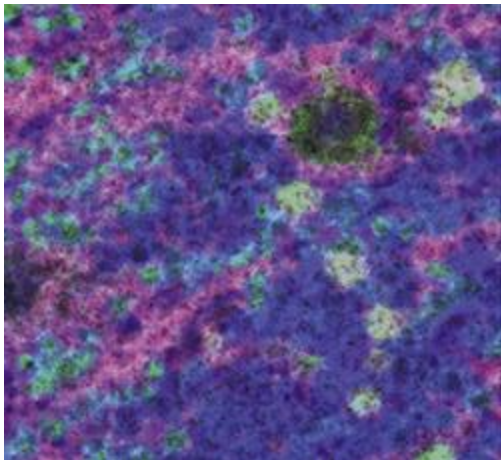
Phenocycler, A.K.A. CODEX



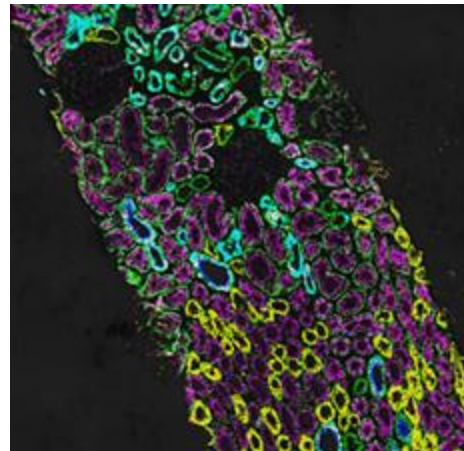
10x Visium



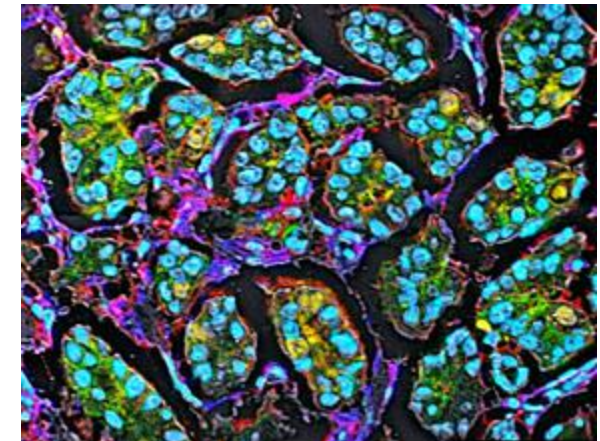
miFISH



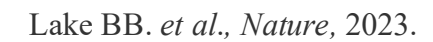
Imaging Mass Spectrometry



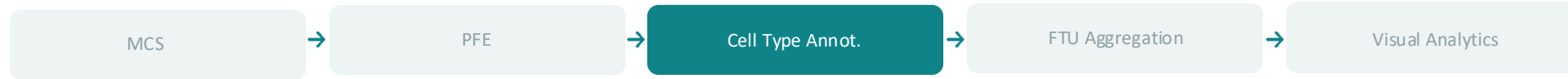
Imaging Mass Cytometry



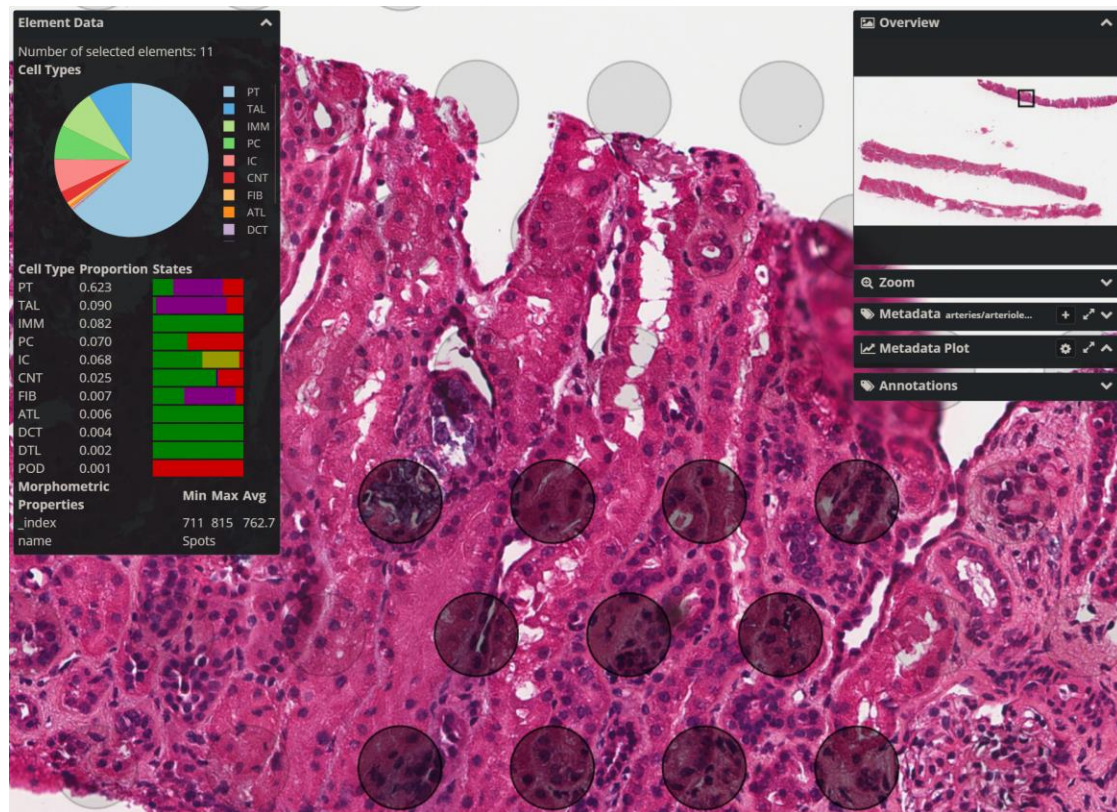
Multiplexed Ion Beam Imaging



Cell Type Annotation: Label Transfer + Spot Annotation



- Label Transfer via Seurat's anchor-based TransferData against a reference (default KPMP kidney reference)
- Deconvolves 55 μm gene expression Visium spots into predicted cell-type proportions



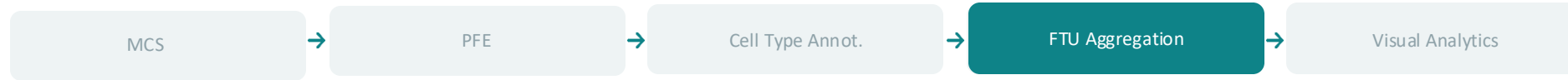
INPUT

Omics feature counts (Seurat rds/Scanpy h5ad) + a cell type reference (e.g. KPMP Kidney Reference)

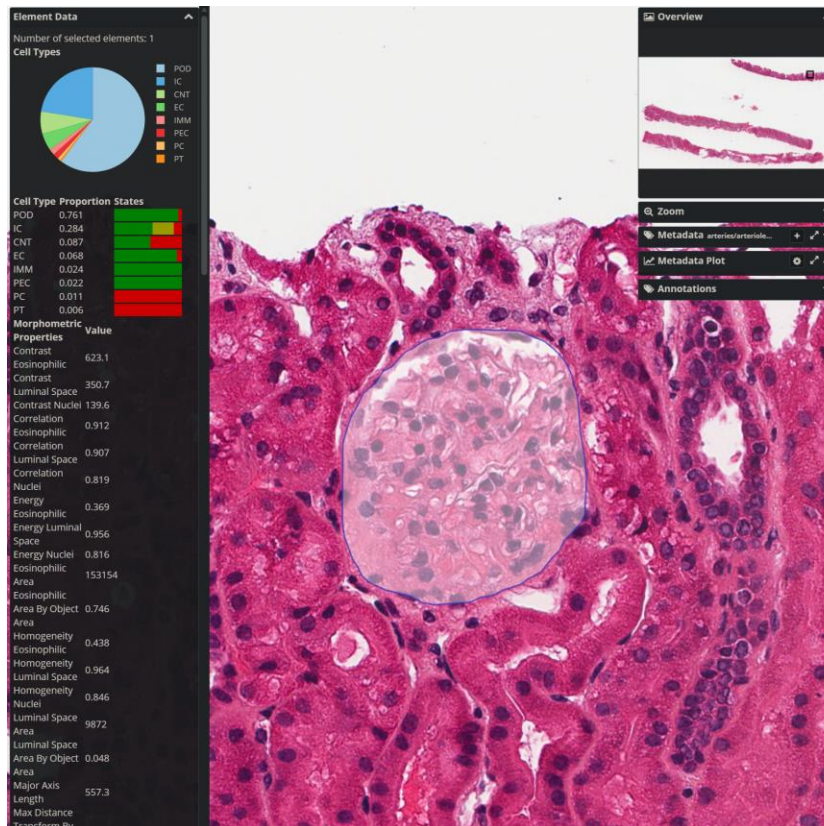
OUTPUT

Every spot annotated with cell type proportions in a downloadable JSON file

Functional Tissue Unit (FTU) Aggregation of Cell Type Annotations



- Overlap-weighted sum of cell-type proportions of all spots at each FTU (Partial spot overlap counts proportionally)
- Merged FTU morphological and molecular signatures integrated in a unified UI for hypothesis generation



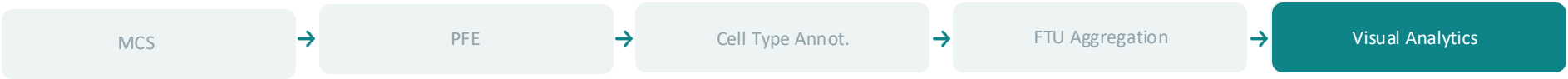
INPUT

Annotated spots + AI-segmented FTU boundaries (JSONs) + FTU Pathomics (csv)

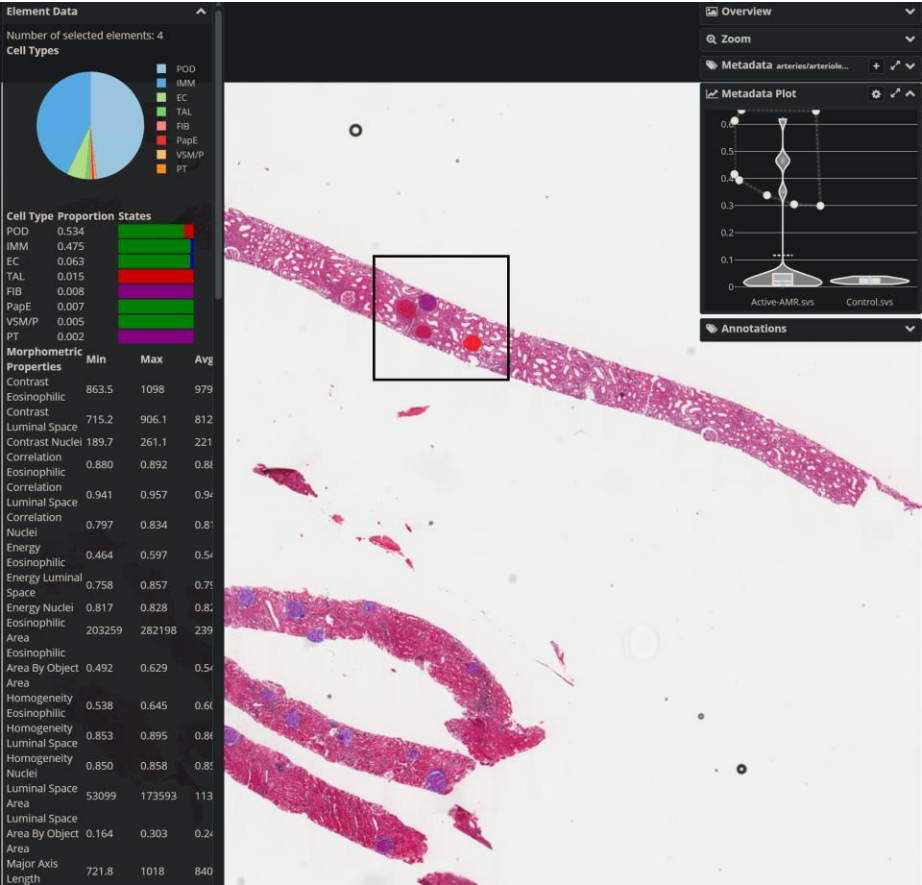
OUTPUT

Per-FTU cell-type composition, joined with pathomic morphometrics (single integrated JSON file)

Visualizing Morphology + Omics in a Unified Interface



Interactive in-tool visualizations to turn FTU-level cell-type composition and pathomic records into discovery



Select & Summarize

Rectangle/lasso-select FTUs or spots on the WSI - live statistical summary of cell-type proportions and pathomic features

Scatter Plots

Correlate any pathomic feature against another, or against cell-type proportions, to surface morphology-omics relationships

Violin Plots

Compare distributions across conditions reveals a continuous injury spectrum across FTUs, beyond a single histological score

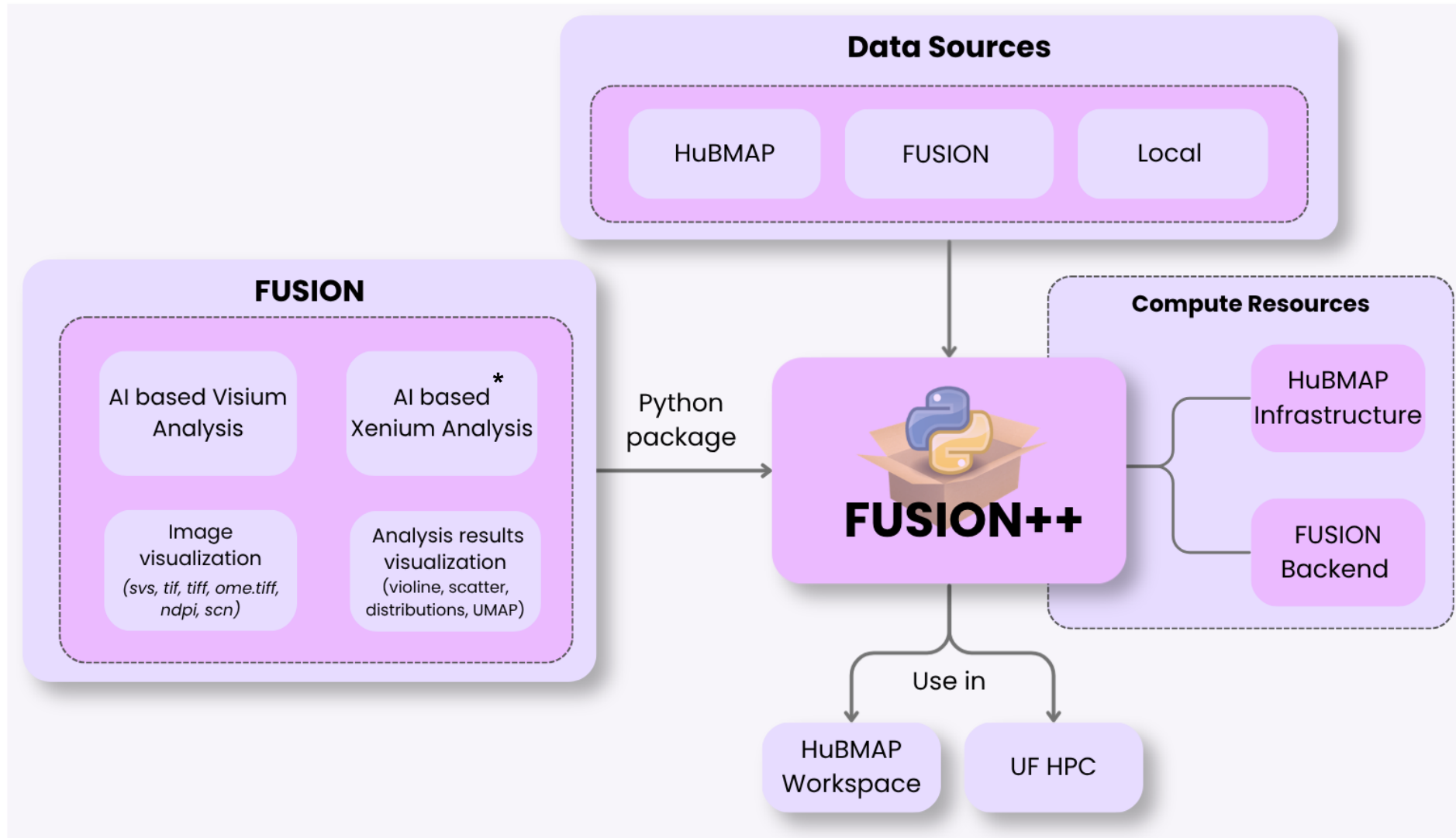
Plot → Slide Lasso

Circle outlier points in a plot; the viewer instantly pans and zooms to those exact structures on the WSI

FUSION++: JupyterHub

Access FUSION from HuBMAP workspace.

FUSION ++ Architecture



* Xenium pipeline integration with FUSION++ in progress

Multi-Source Data Integration

- HuBMAP Consortium
- FUSION Platform
- Local System
- Custom code for other sources
- Globus Transfer for bulk transfers

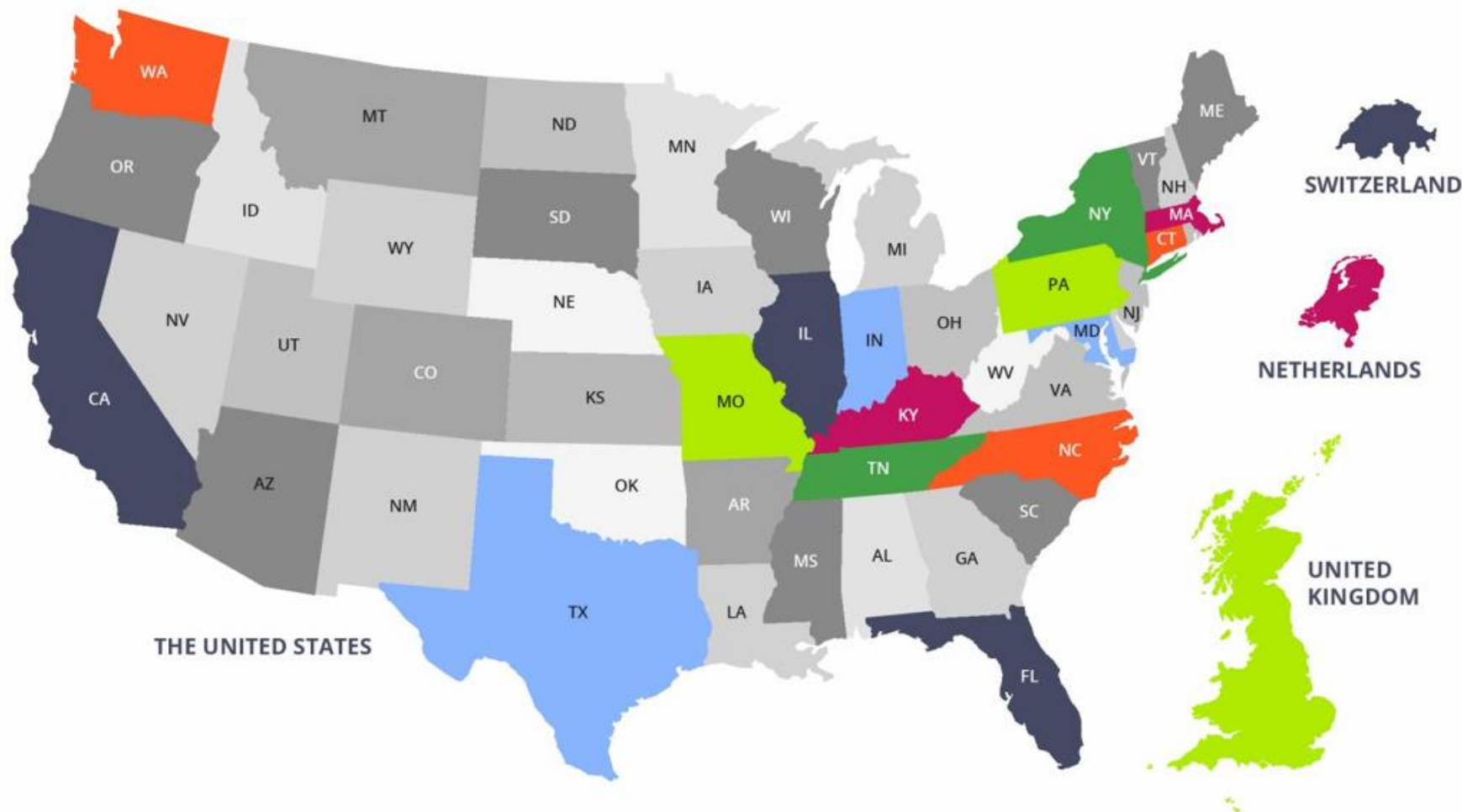
Multi-Source Data Integration

- HuBMAP Consortium
- FUSION Platform
- Local System
- Custom code for other sources
- Globus Transfer for bulk transfers

HuBMAP Consortium

- Human BioMolecular Atlas Program
- Map the human body at single cell resolution.
- Develop tools, resources, and cell atlases to understand cell relationships impact on human health.



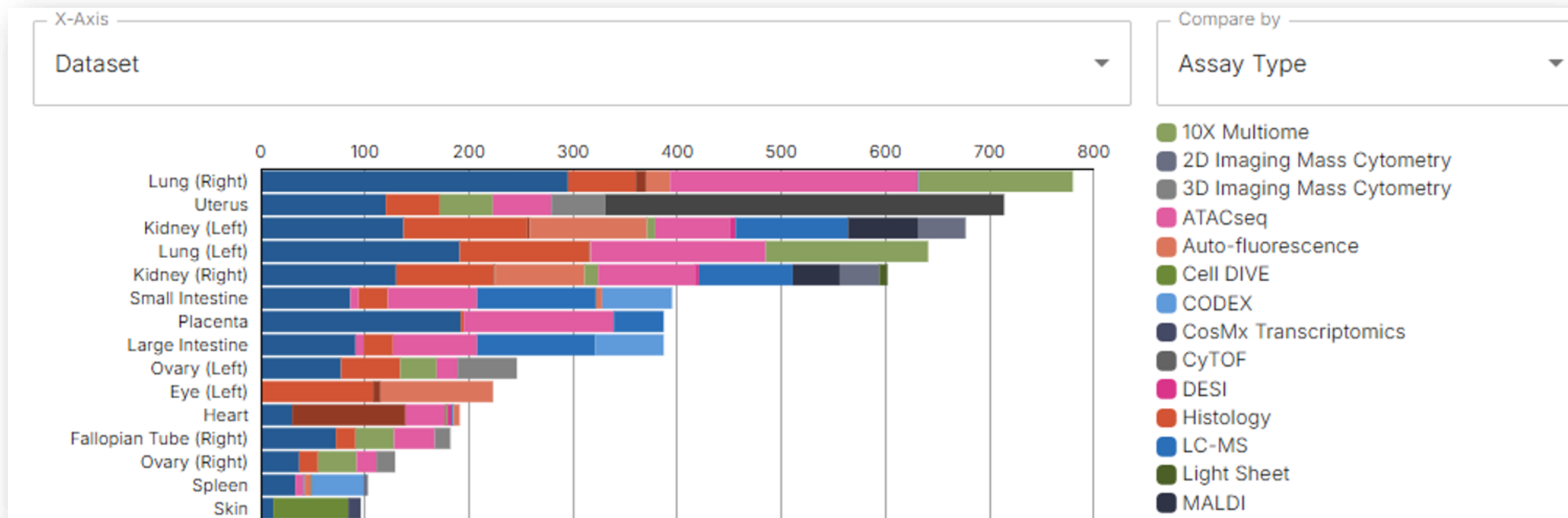


- Beth Israel Deaconess Medical Center
- Brigham and Women's Hospital
- Carnegie Mellon University
- Children's Hospital of Philadelphia
- City of Hope National Medical Center
- Columbia University
- Delft University of Technology
- European Bioinformatics Institute
- General Electric Global Research Center
- Harvard Medical School
- Harvard University
- Indiana University, Bloomington
- IUPUI
- Johns Hopkins University
- Lurie Children's Hospital of Chicago
- National Disease Research Interchange
- New York Genome Center
- NIH Common Fund
- Northwestern University
- Pacific Northwest National Lab
- Pennsylvania State University
- Pittsburgh Supercomputing Center
- Purdue University
- Scripps Research
- Seattle Children's Hospital
- Stanford University
- Texas Advanced Computing Center
- University of California San Diego
- University of California Santa Cruz
- University of Connecticut
- University of Florida
- University of Illinois Chicago
- University of Kentucky
- University of North Carolina at Chapel Hill
- University of Pennsylvania
- University of Pittsburgh
- University of Rochester Medical Center
- University of Zurich
- Vanderbilt University
- Washington University in St. Louis
- Wellcome Sanger Institute
- Yale University

HUBMAP: WHO WE ARE

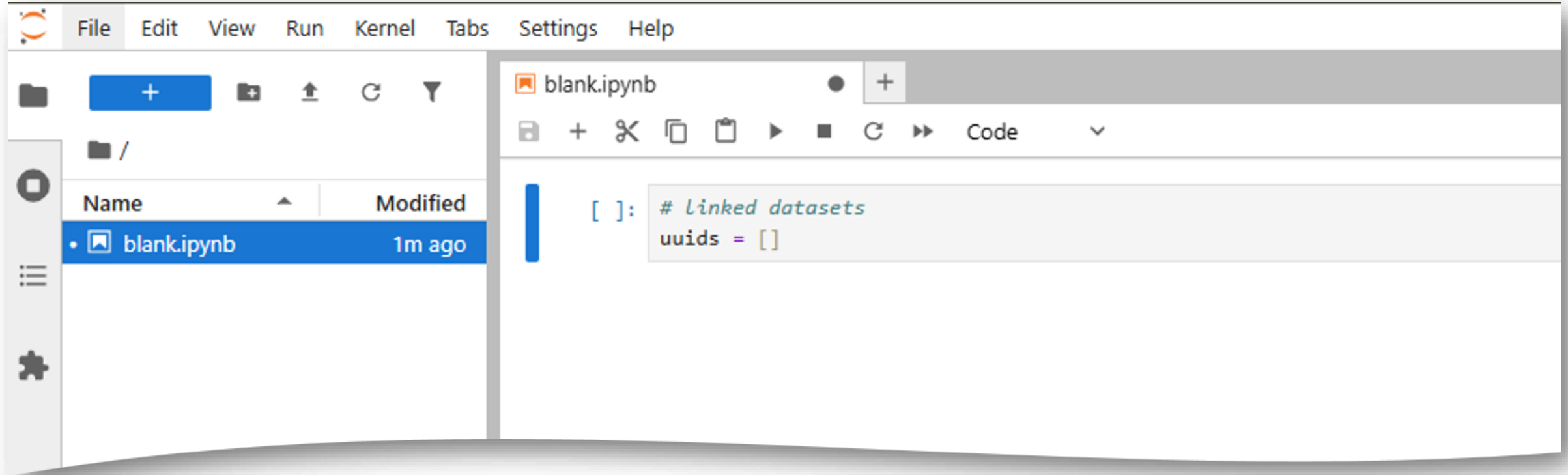
42 Contributing Sites | 15 States | 4 Countries

HuBMAP Data



6,093 Datasets | 31 organs | 20+ assays | 432 donors (male & female, diverse ethnic groups)

HuBMAP Workspace



- A lightweight exploration platform (jupyterHub)
- Access HuBMAP data and perform analyses directly within the portal.

HuBMAP Resources

Time Limit (hours) ⓘ

1 2 3 4 5 6 7 8 9 10 11 12

Memory (GB) ⓘ

8 16 24 32 40 48 56 64 72 80 88 96 104 112 120 128

Number of CPUs ⓘ

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Enable GPU

Disabled ☒ Enabled

*Per individual

HuBMAP Workspace Access

Prerequisite: HuBMAP workspace access – <https://hubmapconsortium.org/workspaces-sign-up-9362/>

The screenshot shows the HuBMAP workspace management interface. At the top, there's a dark blue header with the HuBMAP logo, 'Data' and 'Resources' links, and a user profile icon. Below the header, a notification bar states: 'Manage your workspace collaboration with notifications for invitations shared with you or sent by you. For received invitations, you can preview the workspace details before deciding to accept or decline the invitation.'

Below the notification, there are two tabs: 'Received' (active) and 'Sent'. Under the 'Received' tab, a message box says 'No received workspace invitations.'

The main section is titled 'My Workspaces (2)'. It includes a search bar 'Search workspace by name or ID', a trash icon, a share icon, a 'Stop Workspaces' dropdown, and a '+ Create New' button (highlighted with a red box).

Below the search bar, it shows '0 selected' and two filter buttons: 'Created by Me (2)' (selected) and 'Created by Other (0)'. A table lists the workspaces:

☐	Name ↓	Status ↓	Created By ↓	Last Launched ↓	
☐	new-WSI viz (ID: 1815)	active ✓	Me	2026-01-19	Stop Launch
☐	Jupyter-hub (ID: 1813)	idle ⓘ	Me	2025-12-18	Launch

What's next for FUSION (2.0)

- Integration of single-cell spatial omics (Xenium in-situ).
- Plugin based differential expression, cell neighborhood and niche analysis.
- Agentic AI ecosystem from data ingestion to molecular discovery.
- Broader HPC support beyond workspace (inside institutional boundary).

